

Raspberry Pi Zero W Dual Wi-Fi Setup Guide

Concurrent STA (Client) + AP (Access Point) Mode

This guide demonstrates how to configure a Raspberry Pi Zero W to simultaneously connect to an existing Wi-Fi network (STA mode) while also broadcasting its own access point (AP mode), allowing connected devices to access the internet through the Pi.

Overview

The setup creates two virtual network interfaces:

- **wlan0**: Connects to your home Wi-Fi network (STA mode)
- **uap0**: Virtual AP interface that broadcasts your Pi's own Wi-Fi network

Connected devices → Pi's AP (uap0) → Pi's STA connection (wlan0) → Internet

Prerequisites

- Raspberry Pi Zero W with Wi-Fi capability
- Fresh Raspberry Pi OS Lite (Buster or Bullseye)
- SSH access or monitor + keyboard
- Home Wi-Fi network credentials

Step-by-Step Setup

Step 1: System Preparation

Update your system packages and firmware:

```
bash

sudo apt update && sudo apt full-upgrade -y
sudo reboot
```

Step 2: Install Required Packages

Install the necessary software packages:

```
bash

sudo apt install -y hostapd dnsmasq iw
```

Stop services during configuration:

```
bash
```

```
sudo systemctl stop hostapd  
sudo systemctl stop dnsmasq
```

Package Functions:

- `hostapd`: Creates and manages the wireless access point
- `dnsmasq`: Provides DHCP server functionality for AP clients
- `iw`: Manages wireless network interfaces

Step 3: Configure Wi-Fi Client Connection

Edit the WPA supplicant configuration:

```
bash  
  
sudo nano /etc/wpa_supplicant/wpa_supplicant.conf
```

Add your home Wi-Fi credentials:

```
ini  
  
country=US  
ctrl_interface=DIR=/var/run/wpa_supplicant GROUP=netdev  
update_config=1  
  
network={  
    ssid="YourHomeSSID"  
    psk="YourHomePassword"  
    key_mgmt=WPA-PSK  
}
```

Step 4: Create Virtual AP Interface

Create a virtual access point interface:

```
bash  
  
sudo iw dev wlan0 interface add uap0 type __ap
```

Verify the interface was created:

```
bash  
  
ip a show uap0
```

Step 5: Configure Static IP for AP

Edit the DHCP client daemon configuration:

```
bash  
  
sudo nano /etc/dhpcpd.conf
```

Add at the end of the file:

```
ini  
  
interface uap0  
    static ip_address=192.168.4.1/24  
    nohook wpa_supplicant
```

Restart the DHCP client daemon:

```
bash  
  
sudo service dhpcpd restart
```

Step 6: Configure DHCP Server (dnsmasq)

Backup the original configuration:

```
bash  
  
sudo mv /etc/dnsmasq.conf /etc/dnsmasq.conf.orig
```

Create new dnsmasq configuration:

```
bash  
  
sudo nano /etc/dnsmasq.conf
```

Add the following content:

```
ini  
  
interface=uap0  
dhcp-range=192.168.4.2,192.168.4.20,255.255.255.0,24h
```

Configuration Details:

- `interface=uap0`: Bind to the virtual AP interface
- `dhcp-range`: Assign IPs from 192.168.4.2 to 192.168.4.20

- `(24h)`: 24-hour lease time

Step 7: Configure Access Point (hostapd)

Create the hostapd configuration file:

```
bash  
  
sudo nano /etc/hostapd/hostapd.conf
```

Add the access point settings:

```
ini  
  
interface=uap0  
driver=nl80211  
ssid=PiZeroAP  
hw_mode=g  
channel=7  
wmm_enabled=0  
macaddr_acl=0  
auth_algs=1  
ignore_broadcast_ssid=0  
wpa=2  
wpa_passphrase=YourStrongPassword  
wpa_key_mgmt=WPA-PSK  
rsn_pairwise=CCMP
```

Key Parameters:

- `(ssid)`: Name of your Pi's Wi-Fi network
- `(channel=7)`: Wi-Fi channel (avoid conflicts with your home network)
- `(wpa_passphrase)`: Password for connecting to your Pi's AP

Point hostapd to the configuration file:

```
bash  
  
sudo nano /etc/default/hostapd
```

Uncomment and set:

```
ini  
  
DAEMON_CONF="/etc/hostapd/hostapd.conf"
```

Step 8: Enable IP Forwarding and NAT

Enable IP forwarding permanently:

```
bash
sudo nano /etc/sysctl.conf
```

Uncomment the line:

```
ini
net.ipv4.ip_forward=1
```

Apply immediately:

```
bash
sudo sysctl -w net.ipv4.ip_forward=1
```

Configure Network Address Translation (NAT):

```
bash
sudo iptables -t nat -A POSTROUTING -o wlan0 -j MASQUERADE
sudo iptables -A FORWARD -i wlan0 -o uap0 -m state --state RELATED,ESTABLISHED -j ACCEPT
sudo iptables -A FORWARD -i uap0 -o wlan0 -j ACCEPT
```



Save iptables rules for persistence:

```
bash
sudo sh -c "iptables-save > /etc/iptables.ipv4.nat"
```

Configure automatic rule restoration on boot:

```
bash
sudo nano /etc/rc.local
```

Add the following lines before `exit 0`:

```
bash
```

```
# Create virtual AP interface on boot
iw dev wlan0 interface add uap0 type __ap || true

# Restore iptables rules
iptables-restore < /etc/iptables.ipv4.nat
```

Note: The `|| true` ensures the boot process continues even if the interface already exists.

Step 9: Enable and Start Services

Start the services immediately:

```
bash
```

```
sudo systemctl start hostapd
sudo systemctl start dnsmasq
```

Enable services to start on boot:

```
bash
```

```
sudo systemctl enable hostapd
sudo systemctl enable dnsmasq
```

Step 10: Final Reboot and Testing

Reboot to apply all changes:

```
bash
```

```
sudo reboot
```

Network Architecture



Testing and Verification

After reboot, verify the setup:

1. Check Pi's connection to home Wi-Fi:

```
bash
iwconfig wlan0
```

2. Verify AP interface is up:

```
bash
iwconfig uap0
```

3. Check running services:

```
bash
sudo systemctl status hostapd
sudo systemctl status dnsmasq
```

4. Test connectivity:

- Look for "PiZeroAP" in available Wi-Fi networks on your device
- Connect using the password you set
- You should receive an IP address in the 192.168.4.x range
- Test internet connectivity through the Pi

Network Configuration Summary

Component	Interface	IP Address	Function
Home Wi-Fi	wlan0	DHCP from router	Internet connection
Pi Access Point	uap0	192.168.4.1/24	Local AP gateway
Connected Clients	-	192.168.4.2-20	DHCP assigned

Troubleshooting Tips

- **AP not visible:** Check hostapd service status and configuration
- **No internet on AP clients:** Verify iptables rules and IP forwarding
- **DHCP issues:** Check dnsmasq configuration and service status
- **Interface conflicts:** Ensure uap0 interface is created properly

Use Cases

- **IoT Gateway:** Connect IoT devices that need local network access
- **Portable Hotspot:** Share internet connection in areas with limited access

- **Network Bridge:** Connect devices that can't directly access main network
- **Development Environment:** Isolated network for testing applications

This setup provides a robust dual Wi-Fi configuration that maintains internet connectivity while offering local network services.